

BODMAS

Q.1. Simplify:

$$24 \div 6 \times \left(\frac{3}{4} \text{ of } \frac{1}{3} + \frac{5}{3} \text{ of } \frac{3}{4}\right) = ?$$

- (a) 9 (b) 6 (c) 3 (d) 8

Q.2. Simplify the following. $15 \div 5 \text{ of } 3 \times 4 + 8 \div 12 \text{ of } 4 \times 12 - 4 \div 5 \times 20$

- (a) $\frac{20}{3}$ (b) -10 (c) 12 (d) 1

Q.3. The value of

$$\frac{3 \div 3 \times 2 - 6 \div 3 + 7}{5 \div 5 \times 6 \div 6 + 3 + 3}$$
 is equal to:

- (a) 1 (b) $\frac{3}{2}$ (c) $1\frac{3}{2}$ (d) 2

Q.4. What is the value of R, if,

$$10(8 + 4R) \div (12 \times 5) - 5 \times 2^3 = 6$$

- (a) 86 (b) 68 (c) 76 (d) 67

Q.5. Evaluate $3\frac{1}{2} - [2\frac{1}{4} \div \{1\frac{1}{4} - \frac{3}{4}\}]$

$$(1\frac{1}{2} - \frac{1}{2} - 2\frac{1}{2})$$

- (a) $1\frac{21}{38}$ (b) $2\frac{19}{18}$ (c) $2\frac{21}{38}$ (d) $\frac{47}{38}$

Q.6. Simplify the given expression

$$71 - (-3)\{-2 - (8 - 3)\} + 3\{5 + (-2)(-1)\}$$

- (a) 30 (b) 119 (c) 50 (d) 71

Q.7. Simplify the following expression.

$$\left(\frac{5}{3} + \frac{6}{5} \div \frac{18}{5}\right) \text{ of } \frac{5}{12} - \left(\frac{5}{6} \div \frac{5}{12}\right) \text{ of } \frac{5}{12} + \frac{2}{3} \text{ of } \frac{3}{2}$$

- (a) 1 (b) $\frac{5}{6}$ (c) $\frac{6}{5}$ (d) 0

Q.8. The Value of $[(2^4(20 \times 5 + 10) \div 11) - (2^3 \times 15 \div 20) - (9 \times 16)]$ is:

- (a) 8 (b) 16 (c) 10 (d) 12

Q.9. The value of

$$\frac{3 + 8 \times 8 \div 8 \text{ of } 8 + 7 \div 7 \times 2}{5 \div 5 \text{ of } 4 + 3 \times 3 \div 3 - 3 + 2}$$
 is:

- (a) $1\frac{3}{5}$ (b) $2\frac{2}{3}$ (c) $1\frac{4}{7}$ (d) $2\frac{3}{7}$

Q.10. Simplify the given expression.

$$18 \div 3 \text{ of } 2 \times 5 + 72 \div 18 \text{ of } 2 \times 3 - 4 \div 8 \times 2$$

- (a) 20 (b) $\frac{20}{3}$ (c) $\frac{25}{3}$ (d) 25

Q.11. Simplify the given expression.

$$b + 2a - [(3b + a) - (2a + b) + 2a] - a$$

- (a) -a (b) -b (c) a (d) b

Q.12. If the mathematical symbols in the expression given below are changed from '+' to '÷', '-' to '×', '÷' to '-' and '×' to '+', then what will be the value of the expression?

$$87 \times 256 + 16 - 77 \div 1359$$

- (a) 20 (b) -40 (c) 40 (d) -20

Q.13. If $\sqrt{a} = 3b$, then $\frac{a}{b^2}$ is equal to:

- (a) $\frac{1}{6}$ (b) 6 (c) $\frac{1}{9}$ (d) 9

Q.14. Simplify: $[\frac{1}{5} \text{ of } (35 - 5) + 4]^2 - [9$

$$- \{7 + (2 - 3)\} + 5]^2$$

- (a) 100 (b) 36 (c) 25 (d) 64

Q.15. Simplify the given expression.

$$y + 2x - [y - (y - x + y) - (x + y) + y] - 2y.$$

- (a) -y (b) -2x (c) y (d) 2x

Q.16. Simplify the given expression and find the value for $x = -1$.

$$\frac{10x^2 + 5x + 2xy + y}{5x + y}$$

- (a) -1 (b) 0 (c) 1 (d) 2

Q.17. The value of

$$\left(\frac{2 + 7 \times 7 \div 9 \text{ of } 9 + 6 \div 6 \times 2}{4 \div 4 \text{ of } 5 + 7 \times 7 \div 7 - 6 + 4}\right) \text{ is:}$$

- (a) $\frac{1734}{2407}$ (b) $\frac{1865}{2106}$ (c) $\frac{1857}{2507}$ (d) $\frac{1664}{2305}$

Q.18. What is the value of p, if, $25(4 + 3p) \div 5 - 3 \times 8 = 26$?

- (a) 5 (b) 4 (c) 7 (d) 2

Q.19. Simplify the following expression.

$$\left(\frac{5}{3} + \frac{6}{5} \div \frac{18}{5}\right) \text{ of } \frac{5}{12} - \left(\frac{5}{6} \div \frac{5}{12}\right) \text{ of } \frac{5}{12} + \frac{2}{3} \text{ of } \frac{3}{2}$$

- (a) 1 (b) $\frac{5}{6}$ (c) $\frac{6}{5}$ (d) 0

Q.20. What is the value of

$$\left[\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5}\right] \times \frac{1}{5} - \left[\frac{1}{9} + \frac{1}{9} \times \frac{1}{9}\right] \times \frac{1}{10} + \left(1 - \frac{6}{25}\right)$$

- (a) 2 (b) 1 (c) 3 (d) 4

Q.21. Identify the single rational number that may replace the given expression.

$$[1^2 + 2^2 + 3^2 + 4^2] \times \frac{2}{3 \times 5} \div \left(\frac{1}{4}\right)^2$$

- (a) 16 (b) 64 (c) $\frac{1}{64}$ (d) $\frac{1}{16}$

Q.22. Find the value of the given expression

$$\frac{(4\frac{1}{3} + 3\frac{1}{3} \times 1\frac{4}{5} \div 3\frac{3}{4} \times (1\frac{1}{2} + 1\frac{1}{3}))}{\frac{2}{3} \div \frac{5}{6} \times \frac{2}{3}}$$

- (a) $11\frac{3}{8}$ (b) $10\frac{1}{8}$ (c) $14\frac{3}{8}$ (d) $16\frac{5}{8}$

Q.23. Simplify the following expression

$$\left[\frac{85}{34} \times \frac{1}{18} - \left\{\left(\frac{46}{69} \div \frac{27}{135}\right) - \left(\frac{86}{129} \times \frac{91}{14}\right)\right\}\right] \text{ of } \frac{112}{36}$$

- (a) $3\frac{1}{5}$ (b) $3\frac{3}{4}$ (c) $3\frac{1}{2}$ (d) $3\frac{1}{4}$

Q.24. Evaluate

$$\frac{0.04}{0.05} \text{ of } \frac{(3\frac{1}{3} - 2\frac{1}{2}) \div \frac{1}{2} \text{ of } 1\frac{1}{4}}{\frac{1}{3} + \frac{1}{5} \text{ of } \frac{1}{9}}$$

- (a) 5 (b) 0.3 (c) 3 (d) 0.03

Q.25. If $(48 \div 72 \times 3) - [15 \div 8 \times (40 - 32) - 10] + 2P = 6 \div 2$, then find the value of P

- (a) 2 (b) 4 (c) 1 (d) 3

Q.26. Simplify $\frac{1 - \frac{3}{7} \div \frac{9}{49} \times \frac{3}{14}}{1 + \frac{3}{5} \div \frac{1}{25} \times \frac{1}{30}}$

- (a) $\frac{1}{6}$ (b) $\frac{1}{5}$ (c) $\frac{1}{3}$ (d) $\frac{1}{7}$

Q.27. The value of $1\frac{2}{5} - [3\frac{3}{4} \div (1\frac{1}{4} \div \frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \div 1\frac{1}{3})]$ is:

$$(1\frac{1}{2} \times 3\frac{1}{3} \div 1\frac{1}{3})$$

- (a) 1 (b) 3 (c) 2 (d) 0

Q.28. The value of

$$\frac{1\frac{1}{2} \div 3\frac{1}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5}}{\frac{1}{5} \times 3\frac{1}{2} - \frac{1}{3} \div 1\frac{3}{4} \times 3\frac{1}{2}}$$
 is:

- (a) 38 (b) 33 (c) 36 (d) 40

Q.29. What is the value of $[\frac{1}{8} + \{\frac{1}{6} \times (\frac{36}{45} \div \frac{24}{25}) - (\frac{12}{21} \times \frac{14}{15} \div \frac{24}{45})\} + \frac{27}{36}]$?

- (a) $\frac{1}{36}$ (b) $\frac{1}{72}$ (c) $\frac{1}{108}$ (d) $\frac{1}{27}$

Q.30. The value of $(5 + 3 \div 5 \times 5) \div (3 \div 3 \text{ of } 6) \text{ of } (4 \times 4 \div 4 \text{ of } 4 + 4 \div 4 \times 4)$ is

- (a) $8\frac{1}{5}$ (b) $7\frac{1}{3}$ (c) $9\frac{3}{5}$ (d) $6\frac{2}{3}$

Q.31. The value of $\frac{9}{15} \text{ of } (\frac{2}{3} \div \frac{2}{3} \text{ of } \frac{3}{2}) \div (\frac{3}{4} \times \frac{3}{4} \div \frac{3}{4} \text{ of } \frac{4}{3}) \text{ of } (\frac{5}{4} \div \frac{5}{2} \times \frac{2}{5} \text{ of } \frac{4}{5})$ is:

- (a) $\frac{20}{9}$ (b) $\frac{4}{25}$ (c) $\frac{18}{125}$ (d) $\frac{40}{9}$

Q.32. Find the value

$$\left\{\frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)}\right\}$$

- (a) $\frac{25}{2}$ (b) $\frac{19}{2}$ (c) $\frac{23}{2}$ (d) $\frac{21}{2}$

Q.33. The value of $2\frac{7}{8} \div (3\frac{5}{6} \div \frac{2}{7} \text{ of } 2\frac{1}{3}) \times [(2\frac{6}{7} \text{ of } 4\frac{1}{5} \div \frac{2}{3}) \times \frac{5}{9}]$ is:

- (a) $\frac{1}{4}$ (b) 4 (c) $\frac{1}{23}$ (d) 5

Q.34. The value of $2 \times 3 \div 2$ of $3 \times 2 \div (4 + 4 \times 4 \div 4 \text{ of } 4 - 4 \div 4 \times 4)$ is:

- (a) 8 (b) 1 (c) 4 (d) 2

Q.35. Simplify the following expression.

$$[\sqrt{25} + 12 \div 3 - \{20 + (16 \text{ of } 8 \div 16) -$$

$$(54 \div 18 \text{ of } \frac{1}{2})]$$

- (a) -13 (b) 12 (c) 22 (d) 0

Q.36. Simplify the following

$$\frac{3a+b}{2} - \frac{a-3b}{3} + 2b$$

(a) $\frac{5(a+3b)}{6}$ (b) $\frac{7a+3b}{6}$

(c) $\frac{7(a+3b)}{6}$ (d) $\frac{a+3b}{6}$

Q.37. Simplify the given expression.

$$(a-2b)(b-3a) + (a+b)(a-3b) - (b-3a)(4a-5b)$$

(a) $8a^2 - 10ab$ (b) $8a^2 + 10ab$

(c) $10a^2 - 14ab$ (d) $10a^2 - 4ab$

Q.38. The value of $(3^2)^2 + 3 \times 3 \div 3 - 3$ is:

- (a) 3^2 (b) 9^2 (c) 2^3 (d) 3^0

Q.39. If $720 \div 8 + 915 \div 15 - m + 32 \times 5 = 1104 \div 16 \times 111 \div 37$, then the value of m is:

- (a) 104 (b) 518 (c) 207 (d) 311

Q.40. Simplify the following expression.

$$(4x+1)^2 - (4x+3)(4x-1)$$

- (a) $(4x+1)$ (b) 4 (c) $(4x-3)$ (d) $4x$

Q.41. What value of 'y' will satisfy the following equation?

$$\frac{3y}{1 + \frac{1}{1 + \frac{y}{1-y}}} = 2$$

- (a) $\frac{2}{3}$ (b) $\frac{5}{4}$ (c) $\frac{4}{5}$ (d) $\frac{3}{2}$

Q.42. What is the value of

$$\frac{14 + 7 - 12 + 15 - 17 + 2 - 3 + 17}{2 + 7 - 9 + 2 - 4 + 22} ?$$

- (a) $\frac{23}{20}$ (b) $\frac{17}{20}$ (c) $\frac{19}{20}$ (d) $\frac{33}{20}$

Q.43. The value of

$$(\frac{1}{5}) + [1\frac{1}{4} - \frac{1}{2}] \times (\frac{2}{5})$$
 lies between:

- (a) 2.1 and 2.4 (b) 0.2 and 0.6
(c) 1.5 and 1.8 (d) 0.8 and 1.2

Q.44. The value of $(\frac{1}{2} + \frac{3}{4}) \div (\frac{1}{5} + \frac{3}{10})$ lies between:

- (a) 2.7 and 3.0 (b) 2.0 and 2.2
(c) 1.5 and 1.8 (d) 2.3 and 2.6

Q.45. Evaluate: $5 \times 4 + (3 + 4)^2 \div 7 \times 4$:

- (a) 47 (b) 46 (c) 48 (d) 49

Q.46. The value of $99\frac{95}{99} \times 99 - 95$ is:

- (a) 9897 (b) 9993 (c) 9999 (d) 9801

Q.47. What is the simplified value of $5 \div 10$ of $10 \times 4 + 4 \div 4$ of $4 \times 10 - (10 - 4) \div 16 \times 4$

- (a) 1.2 (b) 2.5 (c) 21 (d) 58.5

Q.48. The simplified value of

$$(3\frac{1}{5} - \frac{3}{5}) \div \frac{8}{5}$$
 is:
$$1\frac{1}{7} \div \{\frac{6}{7} - (\frac{1}{7} \div \frac{1}{5})\}$$

- (a) $\frac{13}{64}$ (b) $\frac{13}{16}$ (c) $\frac{13}{8}$ (d) $\frac{13}{7}$

Q.49. Value of $\frac{8}{9}$ of $(5\frac{1}{4} \div 2\frac{1}{3} \text{ of } 4)$

$$\div (8 \div \frac{2}{3} \text{ of } \frac{4}{5}) \text{ of } (8 \times \frac{2}{3} \div \frac{4}{5})$$
 is

- (a) $1\frac{1}{8}$ (b) $\frac{4}{15}$ (c) $\frac{1}{200}$ (d) $\frac{1}{100}$

Q.50. A student was asked to find the

$$\text{value of } 9\frac{4}{9} \div 11\frac{1}{3} \text{ of } \frac{1}{6} + (1\frac{1}{3} \times 1\frac{4}{5}$$

$$\div \frac{3}{5}) \times 2\frac{1}{6} \text{ of } \frac{2}{3} \div \frac{4}{3} \text{ of } \frac{2}{3}$$
 His

answer was $19\frac{1}{4}$. What is the difference

between his answer and the correct answer?

- (a) $7\frac{3}{4}$ (b) $6\frac{2}{3}$ (c) $7\frac{1}{2}$ (d) $6\frac{1}{3}$

Q.51. The value of $3\frac{5}{6} + [3\frac{2}{3} + \{\frac{15}{4} (5\frac{4}{5} \div 14\frac{1}{2})\}]$ is equal to:

- (a) 9 (b) 6 (c) 7 (d) 8

Q.52. Simplify the following expression:

$$(\frac{3}{4} - \frac{1}{4} \div \frac{1}{4} \text{ of } \frac{2}{5}) \div (\frac{3}{4} \div \frac{2}{3} \text{ of } \frac{3}{5})$$

- (a) $\frac{14}{75}$ (b) $-\frac{70}{27}$ (c) $-\frac{14}{15}$ (d) $\frac{32}{75}$

Q.53. The value of

$$364 - [142 - \{75 + (38 - \frac{5}{4} + \frac{14}{8})\}]$$
 is:

- (a) 332 (b) 330 (c) 331 (d) 334

Q.54. If x is the square of the number when $(\frac{2}{5} \text{ of } 6\frac{1}{4} \div \frac{3}{7})$ of $1\frac{2}{7}$ is divided

by $11\frac{1}{4}$, then the value of $81x$ is:

- (a) 36 (b) 16 (c) 9 (d) 4

Q.55. The value of

$$5\frac{1}{2} \div 3\frac{2}{3} \text{ of } \frac{1}{4} + (5\frac{1}{9} - 7\frac{7}{8} \div 9\frac{9}{20}) \times \frac{9}{11}$$

$$5 \div 5 \text{ of } \frac{1}{10} - 10 \times 10 \div 20$$

is:

- (a) $1\frac{4}{5}$ (b) $1\frac{9}{10}$ (c) $3\frac{4}{5}$ (d) $9\frac{1}{2}$

Q.56. The value of

$$\frac{\frac{1}{3} + [4\frac{3}{4} - (3\frac{1}{6} - 2\frac{1}{3})]}{(\frac{1}{5} \text{ of } \frac{1}{5} \div \frac{1}{5}) \div (\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5})}$$

lies between

- (a) 10.2 and 10.8 (b) 4.2 and 4.4
(c) 8.2 and 8.8 (d) 0.4 and 0.9

Q.57. A student was asked to find the value of

$$(2\frac{1}{3} + 2\frac{1}{2} - \frac{1}{6}) \div 2\frac{1}{3} \times 5\frac{2}{3} \div 1\frac{2}{3} \text{ of } 4\frac{1}{4}$$

$$3\frac{1}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3} + 5\frac{1}{3} \times \frac{3}{4} \div 2\frac{2}{3}$$

His answer was $\frac{6}{7}$. What is the difference between the correct answer and his answer?

- (a) $\frac{9}{14}$ (b) $\frac{5}{14}$ (c) $\frac{11}{49}$ (d) $\frac{6}{49}$

Q.58. The value of $(1\frac{1}{3} + 2\frac{6}{7} \text{ of } 5\frac{3}{5})$

$$\div (6\frac{2}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3}) \times (\frac{3}{4} \times 2\frac{2}{3} \div$$

$$\frac{5}{9} \text{ of } 1\frac{1}{5}) = 1 + k$$
, where k lies

between:

- (a) -0.07 and -0.06 (b) -0.08 and -0.07
(c) -0.06 and -0.05 (d) -0.05 and -0.04

Q.59. The value of $\frac{3}{8} + \frac{1}{7} \times 1\frac{3}{4} +$

$$(\frac{2}{3} \div \frac{3}{10} \text{ of } \frac{4}{9}) - (3\frac{1}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3})$$

$$\times \frac{15}{8}$$
 is:

- (a) $5\frac{1}{2}$ (b) $10\frac{1}{2}$ (c) $10\frac{1}{4}$ (d) $5\frac{3}{8}$

Q.60. What is the simplified value of

$$\{4 - \frac{2}{1 + \frac{2}{1 - \frac{1}{2 + \frac{3}{4}}}}\} \div 1\frac{5}{12} \text{ of } \frac{72}{145}$$

$$(4 + 3 \div 0.5 - 1)$$

- (a) -4 (b) 1 (c) 3 (d) -2

Q.61. The value of $4\frac{2}{5} \div \{(1\frac{1}{2} - 3\frac{1}{5})$

$$\div 3\frac{2}{5} + (2\frac{1}{5} + 1\frac{1}{2} + 4\frac{2}{5}) + \frac{1}{2}\} = ?$$

- (a) $\frac{4}{7}$ (b) $\frac{3}{7}$ (c) $\frac{3}{5}$ (d) $\frac{3}{4}$

Q.62. Simplify the following expression

$$3\frac{6}{7} - \{5\frac{1}{7} - (2\frac{3}{7} - 3\frac{5}{7} - 4)\}$$

- (a) $\frac{34}{7}$ (b) $\frac{10}{7}$ (c) $\frac{20}{7}$ (d) $\frac{44}{7}$

Q.63. $\{(28 - 3^3) + 99\} + 136 \div 4 \times 5 - (44 - 2^2) = ?$

- (a) 250 (b) 200 (c) 230 (d) 270